Team Details

Team Name: Silli-Con Artists

SR. NO	ROLE	NAME	ACADEMIC YEAR
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Problem Statement Addressed

Drift-Sense — Navigation-Error Recovery (Applied Materials, PS 02)

WHY NAVIGATION-ERROR RECOVERY MATTERS

A wafer-inspection tool navigates to a defect by stage coordinates. Stage drift means the field it actually images is not the field it was sent to. Recovery means re-locating a known high-magnification reference pattern inside a wider, coarser, noisier search image.

The reference is 1000×1000 px at 1 nm/px; the search covers 10× the area at 10 nm/px, so the reference occupies only 100×100 px of it. DRAM and FinFET layouts are periodic by design, so a wrong repeat is not a blurry near-miss — it is a structurally valid match that looks correct. That ambiguity, not similarity, is the real problem.

10×

magnification gap between reference and search

100×100

pixels the reference occupies in the search field

~163

lattice cells inside one reference — every one a candidate

Getting it wrong is a data-integrity failure, not an accuracy one: the tool reports a confident measurement of the wrong cell.

Idea Description

DRAM-style layouts · classical, physics-based, zero-training localization

KEY CONCEPT

We do not match images.
We invert the microscope.

The forward model is known — beam PSF, area-average decimation, geometric warp, Poisson shot noise then Gaussian read noise. Localization is therefore a maximum-likelihood inverse problem with a few nuisance parameters, not a similarity search.

WHY IT BEATS TEMPLATE MATCHING

The periodic lattice everyone treats as the enemy is used as a ruler: it fixes pose. The aperiodic residual — line-placement jitter unique to each cell — supplies identity.

The baseline resizes over a few fixed scales and takes an argmax, so it cannot represent rotation or off-nominal magnification, and has no way to tell two identical repeats apart.

Proposed Solution

Generator, localizer, and the pipeline from image pair to (x, y)

GENERATOR — 30% of the score, every stage literature-justified

vector DRAM/FinFET layout at 1 nm/px → SE edge brightening (yield rises with surface tilt — Villarrubia, NIST) → beam PSF → rotation 0–2° and scale 9–11:1 → raster shear and jitter → barrel → Poisson(dose) → Gaussian(read) → gamma, vignette, charging streaks.

Independent RNG streams per capture, so the pair is two acquisitions of one scene rather than one image plus noise. Fractional crop origins give continuous ground truth — which is what makes a sub-pixel claim measurable at all.

LOCALIZER — classical, deterministic, CPU-only

measure pose by pyramid search → build the template through the exact forward operator (area-average decimation is the physically correct downsampler, not an approximation) → ZNCC candidate surface → lattice-residual proposals → cheap screen ranks the field →

survivors get a dense 5×5 pose refit → read each candidate's pose grid as EVIDENCE rather than as its maximum → upsampled-DFT sub-pixel → bounded raster-drift correction.

Innovation and Uniqueness

What is genuinely new, and what the measurements say about it

KEY INNOVATION

Read the pose grid as EVIDENCE, not as its maximum.

The refit scores each candidate over ~25 poses. Taking the best of 25 is upward-biased, and the bias grows with how rough that candidate's surface is — so one lucky sample outranks a candidate that was consistently good. Summarising the same grid by a log-sum-exp costs zero extra correlations and is the largest single gain in the project.

Six re-ranking criteria were built and all six failed. The only selection stage that ever worked re-scores by the SAME criterion at a better geometry.

MEASURED ADVANTAGE

95.0%

located within 5 px

55 / 0

fixed / broken vs baseline

Mis-lock on the same 100 pairs: 0.0% / 13.3% / 3.3% against the baseline's 25.0% / 76.7% / 90.0% — strictly dominant, tested as a paired comparison on identical pairs rather than two averages compared by eye.

The baseline's output is always integer + 50.0, so it carries a half-pixel quantisation floor and cannot score on the sub-pixel rung of the tolerance ladder at all.

Results

100 evaluated pairs across three splits — one a held-out architecture

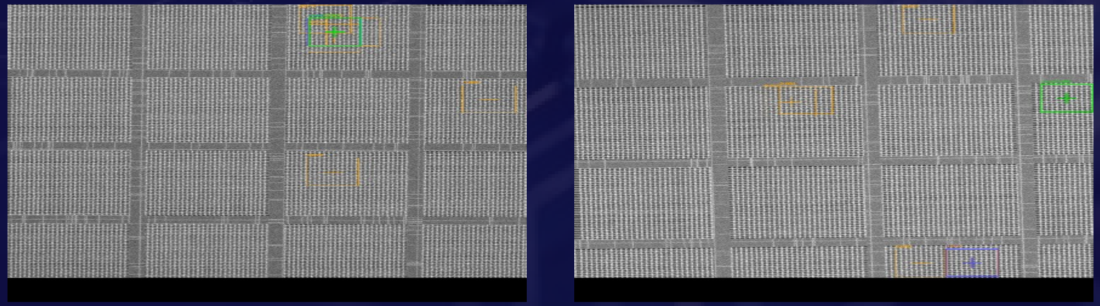
ACCURACY vs THE TOLERANCE LADDER

	sponsor	bench	FinFET
≤ 5 px	100.0%	86.7%	96.7%
≤ 2 px	100.0%	83.3%	93.3%
≤ 1 px	97.5%	83.3%	90.0%
≤ 0.5 px	92.5%	66.7%	76.7%
median err	0.179	0.300	0.214

5 of 100 pairs
~0.6 s per pair, CPU only
baseline sub-pixel: 17.5%

5.0%
aggregate mis-lock

ONE SUCCESS, ONE HONEST FAILURE



SUCCESS — error 0.04 px
prediction sits on the truth

FAILURE — error 628.24 px
a valid lattice repeat won

green prediction · blue truth · orange rivals — search image shown

Technology & Feasibility/Methodology Used

CPU-only · eight pinned dependencies · no network access, no model download

STACK, HARDWARE AND FEASIBILITY

Python 3.14 · numpy · scipy · opencv-python-headless · scikit-image · pillow · pandas · PyYAML · matplotlib, every version pinned. torch is optional and lazily imported — uninstalling it leaves everything working.

Hardware: one laptop CPU. No GPU, no cloud, nothing downloaded at runtime. Deterministic: a single seeded numpy Generator is threaded through, and a test asserts the same seed gives byte-identical images and identical predictions on the same platform.

Inference accepts .png and .npy pairs identically; a separate documented converter exists for visual inspection but is never on the scoring path.

Model size: none. Nothing is trained, so there are no weights to ship, load or version.

Dataset generation ~2 s per pair (a 10 000×10 000 canvas at 1 nm/px); localization ~0.6 s per 1000×1000 pair.

Software Architecture

Hardware Components

Development Tools

GitHub & Video Link

GitHub Repository

https://github.com/Cosmic-dorito/driftlock

Prototype / Simulation Video

Clone, generate a pair, and localize it — three commands in the README

References

VERIFIED REFERENCES — full table in docs/REFERENCES.md

Mäkitalo & Foi. Optimal Inversion of the Generalized Anscombe Transformation for Poisson–Gaussian Noise. IEEE TIP 22(1):91–103, 2013. doi:10.1109/TIP.2012.2202675

Guizar-Sicairos, Thurman & Fienup. Efficient subpixel image registration algorithms. Optics Letters 33(2):156–158, 2008. doi:10.1364/OL.33.000156

Villarrubia, Ritchie & Lowney. Monte Carlo modeling of secondary electron imaging in three dimensions. Proc. SPIE 6518, 65180K, 2007. doi:10.1117/12.712353

Sugiyama et al. Pattern matching method and computer program for executing pattern matching. US Patent 7,925,095 B2, Hitachi High-Technologies, 2011

Abe et al. Image processing method for determining matching position between template and search image. US Patent 8,139,868 B2, Hitachi High-Technologies, 2012

$6F^2$ DRAM cell design with 3F-pitch folded digitline sense amplifier. US Patent 7,349,232 B2

Reddy & Chatterji. An FFT-Based Technique for Translation, Rotation and Scale-Invariant Image Registration. IEEE TIP 5(8):1266–1271, 1996. doi:10.1109/83.506761

Kovesi. Phase congruency: A low-level image invariant. Psychological Research 64(2):136–148, 2000. doi:10.1007/s004260000024

Rule R1 — a citation without a verified DOI/URL, a named verifier and a date is not a citation and does not appear here.

The two Hitachi patents matter twice over: they establish that a closest-to-origin tie-break for periodic SEM patterns is industrial practice, not an arbitrary rule.